

SSTO fusion powered spaceplane using CHARM architecture with $p\text{-}^{11}\text{B}$ fuel

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Abstract

We specify a single-stage-to-orbit (SSTO) spaceplane that flies Space Shuttle-style operations—including a Shuttle-class cargo bay—from a municipal airport to International Space Station (ISS) altitude in low Earth orbit (LEO), powered by a continuous Chambered Aneutronic Rotating Mirror (CHARM) $p\text{-}^{11}\text{B}$ plant [1] with direct energy conversion (DEC). Flight regimes use free air, then scarce air, then carried water. Each design step is written as a closed set of equations. We guesstimate the reactor mass hole, constrain water as a function of dry mass and vacuum Δv , impose a 1 GW plant with space restart and DEC, and solve a reference all-up mass. Combined-cycle engine maps and CHARM size/performance constraints follow.

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1 Vehicle vision

Municipal runway to ISS-class LEO: a Shuttle-style SSTO spaceplane with a real cargo bay, a single-deck crew module, and a continuous CHARM $p\text{-}^{11}\text{B}$ plant driving a combined-cycle engine (free air $\rightarrow$ scarce air $\rightarrow$ carried water). The figures below are the vehicle picture; the equations that close the mass and energy budgets follow.

1.1 Interior floorplan and exterior profile

Crew volume flattens the **Space Shuttle crew module** from two decks to **one** [14,21], then **stretches** the pressurized nose so life support and a suited airlock are not cartoon-thin. Reference overall length $L \approx 52$ m. ECLSS = **Environmental Control and Life Support System**. Depth $\approx 6.5\text{--}7$ m, span ≈ 28 m.

Figures 1 and 2 are orthographic CAD views of one station map (nose left, length 52 m): crew/ECLSS 0–11 m (flight deck + six seats, internal O_2/N_2 , port **ground-only** side hatch); airlock 11–15 m (hatches cabin $\leftrightarrow$ airlock and airlock $\leftrightarrow$ bay only—no outer side door); cargo bay 15–33.3 m (18.3×4.6 m, top clamshell payload-bay doors); flight battery 33.3–35.5 m; $p\text{-}^{11}\text{B}$ fuel 35.5–37.5 m; CHARM 37.5–45 m; water 45–49 m; combined-cycle engine 49–52 m. The floorplan is a top-down cutaway (no landing gear). The profile shows white upper OML, dark TPS belly, extended gear, the port crew hatch, and closed top bay doors. Both views are driven from the same geometry so stations stay consistent if the layout expands.

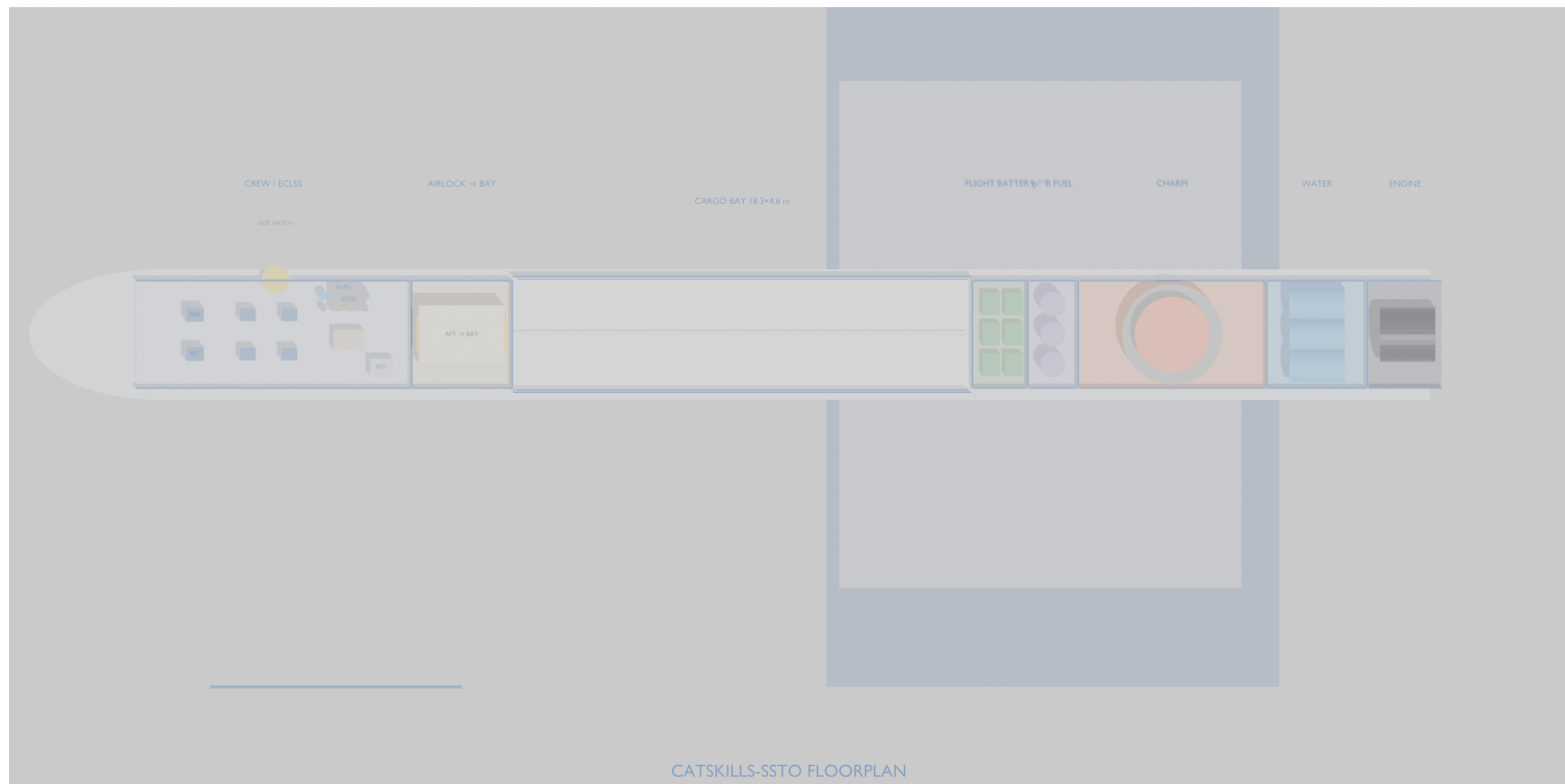


Figure 1 Vehicle floor plan.



Figure 2 Vehicle profile view.

1.2 CHARM power plant

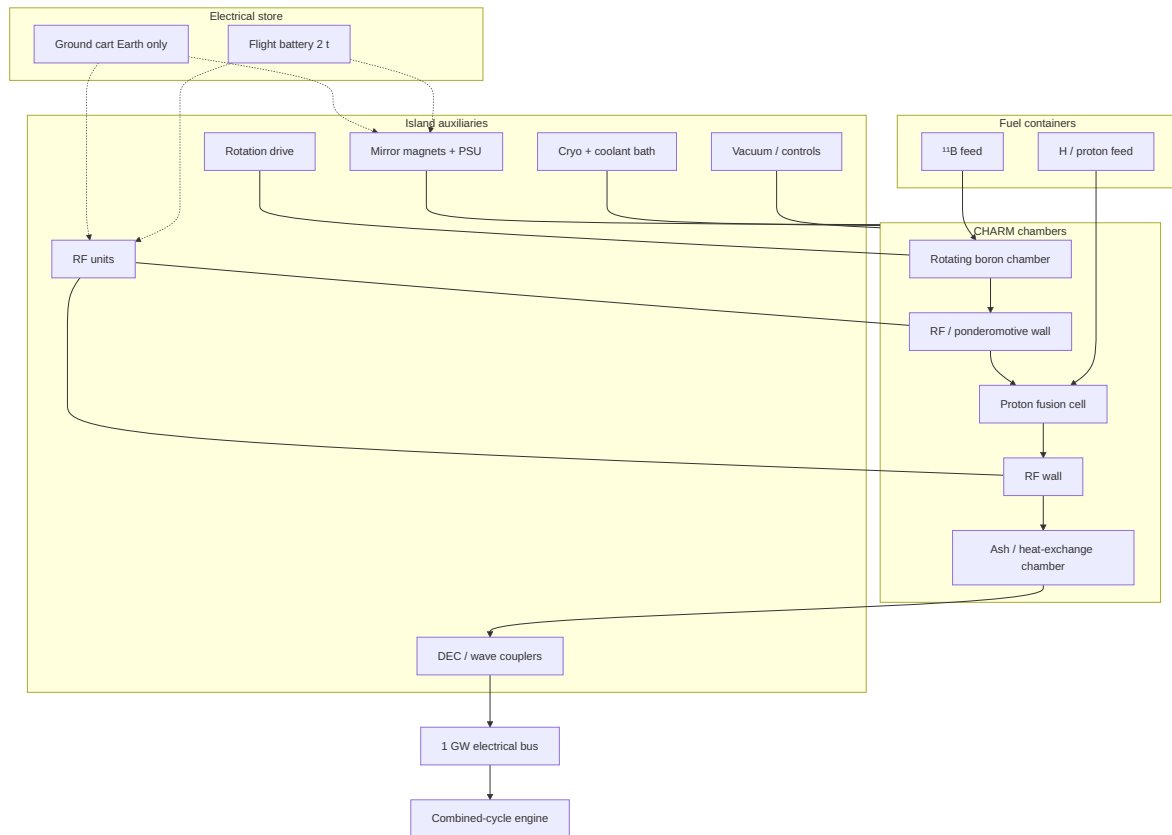


Figure 3 CHARM power-plant schematic.

1.3 Profile stations

2 Design goals (the plane)

Table 1 Design goals for the plane.

ID	Goal	Statement
G1	SSTO	Single stage from runway to ISS-class LEO; no discarded boosters or external tank
G2	Shuttle style	Orbiter-like airframe: wing-body, thermal protection system (TPS), runway landing, gear, control surfaces; crew systems (toilet, ECLSS, food)
G2b	Single-deck crew	Flatten Shuttle cabin to one deck: flight deck; six reclining seats; O ₂ /N ₂ + ECLSS; luggage; forward/side ground door; large aft airlock into bay

ID	Goal	Statement
G2c	Length	Stretch OML as needed for airlock, ECLSS tanks, battery, and fusion-fuel tanks—reference $L \approx 52$ m
G3	Shuttle cargo bay	Usable bay $\approx 18.3 \text{ m} \times 4.6 \text{ m}$ class for payload—not filled with reactors
G4	Payload	Shuttle-class cargo: $m_{\text{pl}} = 24\,400 \text{ kg}$ reference
G5	Destination	Circular LEO compatible with ISS altitude ($\approx 400 \text{ km}$); plane-change to 51.6° treated as margin
G6	Municipal airport	Takeoff/landing on long civil runways; first gear cool enough for noise/oversight; no vertical pad
G7	One engine	Single combined-cycle propulsion string; deadstick glide if plant fails
G8	Clean fuel	$p + {}^{11}\text{B} \rightarrow 3\alpha + 8.7 \text{ MeV}$; CHARM bottle; DEC-first electricity
G9	Power	Plant electrical bus peak $P_\star = 1 \text{ GW}$ (design target)

Not goals: expendable stages; D–T breeding plant; filling the bay with the fusion island.

3 Constants and symbols

Table 2 Physical constants and reference symbols.

Symbol	Meaning	Reference value
R_E	Earth radius	$6.371 \times 10^6 \text{ m}$
μ_E	GM_E	$3.986 \times 10^{14} \text{ m}^3/\text{s}^2$
h_{ISS}	ISS-class altitude	$4.00 \times 10^5 \text{ m}$
r	Orbit radius	$R_E + h_{\text{ISS}}$
g_0	Standard gravity	9.80665 m/s^2
m_{pl}	Cargo-bay payload	$2.44 \times 10^4 \text{ kg}$
$P_\star$	CHARM bus peak power	$1.00 \times 10^9 \text{ W}$

Masses (all kg):

Table 3 Mass symbol definitions.

Symbol	Meaning
m_{af}	Airframe primary: fuselage, wings, TPS (excl. gear/controls)
m_{gear}	Landing gear (nose + mains), doors, actuators
m_{ctrl}	Control surfaces + actuators (elevons, rudder, speed brake / body flap equiv.)

Symbol	Meaning
m_{crew}	Crew cabin systems: ECLSS, O ₂ /N ₂ tanks, pressure control, toilet, galley, food, luggage, airlock fittings
m_{str}	$m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}$
m_{eng}	Combined-cycle engine + inlets + nozzles + ducts
m_{C}	CHARM island (chambers, magnets, radio-frequency (RF), DEC, local shield, cryo, bath)
m_{bat}	Flight battery / auxiliary power unit (APU) (restart + hotel)
m_{f}	p - ¹¹ B fuel
m_{w}	Water carried at takeoff (vacuum reaction mass)
m_{dry}	All mass except water
m_0	Gross liftoff mass (GLOW)
m_{ins}	Mass at LEO insertion (after water burn)

4 MWh budget: Shuttle-class mass to ISS LEO

4.1 Ideal orbital specific energy

$$r = R_E + h_{\text{ISS}}, \quad v_{\text{orb}} = \sqrt{\frac{\mu_E}{r}}, \quad \varepsilon_{\text{orb}} = -\frac{\mu_E}{2r}, \quad \varepsilon_{\text{surf}} \approx -\frac{\mu_E}{R_E}.$$

$$\Delta\varepsilon = \varepsilon_{\text{orb}} - \varepsilon_{\text{surf}} = \frac{\mu_E}{R_E} - \frac{\mu_E}{2r} \approx 3.31 \times 10^7 \text{ J/kg} = 33.1 \text{ MJ/kg}.$$

Numerical anchors:

$$v_{\text{orb}} \approx 7.67 \text{ km/s}, \quad \Delta\varepsilon \approx 9.19 \text{ kWh/kg}.$$

4.2 Orbital energy of the inserted vehicle

Everything that arrives at ISS altitude is still aboard (SSTO):

$$E_{\text{orb}} = m_{\text{ins}} \Delta\varepsilon.$$

In MWh:

$$E_{\text{orb}}^{(\text{MWh})} = m_{\text{ins}} \cdot \frac{\Delta\varepsilon}{3.6 \times 10^9}.$$

Table 4 Orbital energy for representative inserted masses.

m_{ins}	E_{orb}	E_{orb}
100 t (orbiter+cargo class)	3.31 TJ	920 MWh
150 t	4.97 TJ	1380 MWh
200 t	6.62 TJ	1840 MWh

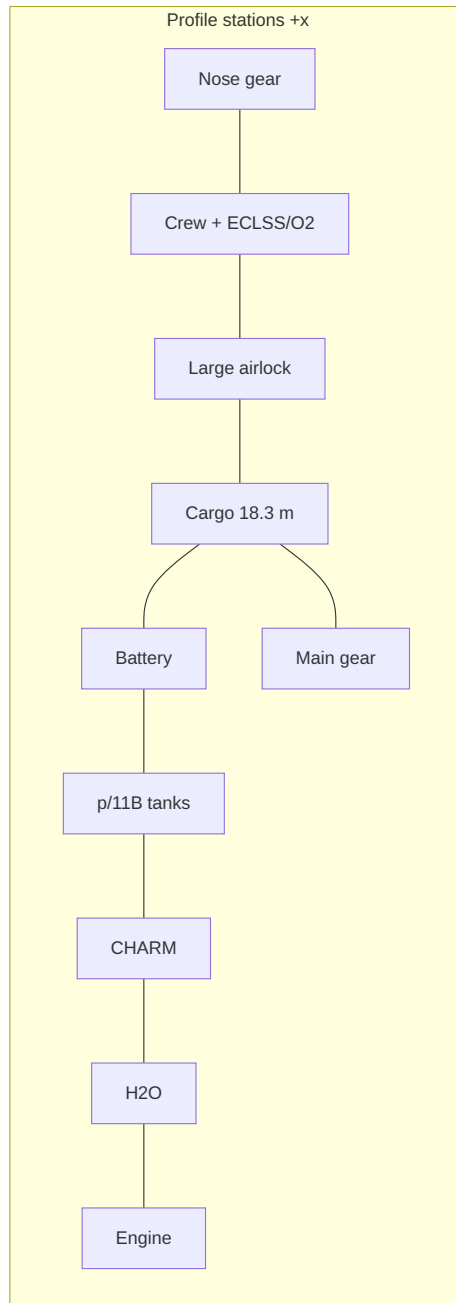


Figure 4 Profile view (to scale).

4.3 Source energy (plant must supply)

Air-breathing and rocket/plasma paths leave energy in the wake and fight drag/gravity. Define a mission multiplier $\kappa_E \geq 1$:

$$E_{\text{src}} = \kappa_E E_{\text{orb}} = \kappa_E m_{\text{ins}} \Delta \varepsilon.$$

Working band for planning:

$$\kappa_E \in [2, 4] \Rightarrow E_{\text{src}}^{(\text{MWh})} \approx (18\text{--}37) m_{\text{ins},100\text{t}}$$

i.e. about 1.8–3.7 GWh of source energy per 100 t inserted, scaling linearly with m_{ins} .

Time at constant bus power $P_\star$:

$$t_{\text{src}} = \frac{E_{\text{src}}}{P_\star} = \frac{\kappa_E m_{\text{ins}} \Delta \varepsilon}{P_\star}.$$

For $m_{\text{ins}} = 1.9 \times 10^5 \text{ kg}$, $\kappa_E = 3$, $P_\star = 1 \text{ GW}$:

$$E_{\text{src}} \approx 18.9 \text{ TJ} \approx 5240 \text{ MWh}, \quad t_{\text{src}} \approx 5.2 \text{ h}.$$

So a 1 GW CHARM bus is an **energy-throughput** match for Shuttle-class SSTO only if the climb/insert lasts hours at high average power—or peak $P_\star$ is used with κ_E toward the low end and a lighter m_{ins} .

ISS note. Matching ISS inclination adds plane-change Δv . Fold into vacuum Δv margin (§5) rather than into $\Delta \varepsilon$.

5 Flight regimes: free air $\rightarrow$ scarce air $\rightarrow$ water

Define cutoff speed v_{ab} and altitude h_{ab} where intakes seal and water thrust begins.

Table 5 Flight regimes and reaction-mass path.

Regime	Density	Reaction mass	Plant $\rightarrow$ thrust path
R1 Free air	Troposphere / lower stratosphere	Free ingested air	Electricity $\rightarrow$ fan / bypass
R2 Scarce air	Upper atmosphere	Free ram air	Electricity + heat $\rightarrow$ ram / plasma accelerator
R3 Vacuum	Exo-atmosphere / LEO insert	Carried water	Electricity / DEC $\rightarrow$ water plasma jet

Energy split (schematic):

$$E_{\text{src}} = E_{\text{R1}} + E_{\text{R2}} + E_{\text{R3}} + E_{\text{hotel}} + E_{\text{loss}}.$$

With free reaction mass in R1–R2, plant energy primarily raises vehicle mechanical energy and pays drag:

$$P_{\text{prop}}(t) \approx \frac{D(t) v(t)}{\eta_p(t)}, \quad \int P_{\text{prop}} dt \subset E_{\text{R1}} + E_{\text{R2}}.$$

In R3, jet power for exhaust speed v_e and mass flow $\dot{m}_w$ is

$$P_{\text{jet}} = \frac{1}{2} \dot{m}_w v_e^2, \quad T = \dot{m}_w v_e \quad \Rightarrow \quad P_{\text{jet}} = \frac{1}{2} T v_e.$$

6 Water mass as a function of whole dry mass

Water is used only in R3. Let

$$m_{\text{dry}} = m_{\text{str}} + m_{\text{pl}} + m_{\text{eng}} + m_{\text{C}} + m_{\text{bat}} + m_{\text{f}},$$

$$m_0 = m_{\text{dry}} + m_{\text{w}}, \quad m_{\text{ins}} = m_{\text{dry}}$$

(all water expended by insertion). Rocket equation for vacuum Δv_{vac} :

$$\Delta v_{\text{vac}} = v_e \ln \left(\frac{m_{\text{dry}} + m_{\text{w}}}{m_{\text{dry}}} \right) = v_e \ln \mu, \quad \mu = e^{\Delta v_{\text{vac}}/v_e}.$$

Water constraint (closed form):

$$m_{\text{w}} = m_{\text{dry}} \left(e^{\Delta v_{\text{vac}}/v_e} - 1 \right) = m_{\text{dry}} (\mu - 1)$$

$$m_0 = m_{\text{dry}} \mu = m_{\text{dry}} e^{\Delta v_{\text{vac}}/v_e}$$

Vacuum Δv budget:

$$\Delta v_{\text{vac}} = v_{\text{orb}} - v_{\text{ab}} + \Delta v_{g,\text{vac}} + \Delta v_{\text{steer}} + \Delta v_{\text{ISS plane}}.$$

Reference planning values:

$$v_{\text{ab}} = 3.5 \text{ km/s}, \quad \Delta v_{g,\text{vac}} + \Delta v_{\text{steer}} = 0.8 \text{ km/s}, \quad \Delta v_{\text{ISS plane}} = 0.2 \text{ km/s} \Rightarrow \Delta v_{\text{vac}} \approx 5.2 \text{ km/s}.$$

Optimistic air-breathing ($v_{\text{ab}} = 5 \text{ km/s}$, small margins) $\rightarrow \Delta v_{\text{vac}} \approx 3.5\text{--}4 \text{ km/s}$.

Exhaust speed from specific impulse:

$$v_e = I_{\text{sp}} g_0.$$

Table 6 Water mass fraction versus vacuum Δv and I_{sp} .

Δv_{vac}	I_{sp}	v_e	μ	$m_{\text{w}}/m_{\text{dry}}$
4 km/s	2000 s	19.6 km/s	1.226	0.226
4 km/s	3000 s	29.4 km/s	1.146	0.146
5.2 km/s	2000 s	19.6 km/s	1.302	0.302
5.2 km/s	5000 s	49.0 km/s	1.112	0.112

7 Vehicle sizing equations

7.1 Structure, crew, gear, controls, and bay

Cargo bay geometry (goal G3):

$$L_{\text{bay}} = 18.3 \text{ m}, \quad D_{\text{bay}} = 4.6 \text{ m}, \quad V_{\text{bay}} \approx \frac{\pi}{4} D_{\text{bay}}^2 L_{\text{bay}} \approx 304 \text{ m}^3.$$

Payload density check:

$$\bar{\rho}_{\text{pl}} = \frac{m_{\text{pl}}}{V_{\text{bay}}} \approx 80 \text{ kg/m}^3$$

(compatible with mixed cargo; bay remains payload volume).

Crew (Shuttle functions, single deck). Forward **flight deck**: commander and pilot **facing forward** into windows and a full control-panel wall [14,21]. Living volume: **six seats** with moderate nap recline; **waste collection system (WCS)**; **galley/food**; **crew luggage / personal stowage**; **ECLSS** with explicit **oxygen and nitrogen / pressure-control tankage** (not a token rack). **Forward/port crew door (side hatch)** for terrestrial ingress only. **Airlock** oversized vs a suit-closet: dual-hatch volume on the aft cabin bulkhead facing the **cargo bay**, sized for suited egress (Shuttle middeck airlock pattern, not undersized) [21].

Landing gear and control surfaces are explicit mass lines (not buried only in a lump “structure” number):

$$\begin{aligned} m_{\text{af}} &= 7.20 \times 10^4 \text{ kg} && \text{(longer fuselage, wings, TPS, doors),} \\ m_{\text{gear}} &= 4.00 \times 10^3 \text{ kg} && \text{(nose + dual main trucks; mass bill only—see profile),} \\ m_{\text{ctrl}} &= 3.00 \times 10^3 \text{ kg} && \text{(elevons, rudder, speed-brake equiv. + actuators),} \\ m_{\text{crew}} &= 8.50 \times 10^3 \text{ kg} && \text{(ECLSS, O}_2\text{/N}_2\text{, WCS, galley, food, luggage, airlock),} \\ m_{\text{str}} &= m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}} = 8.75 \times 10^4 \text{ kg.} \end{aligned}$$

CHARM and propulsion water **do not** consume V_{bay} . Plant volume sits in a **fuselage island** (spine / aft of bay / wing carry-through), subject to

$$V_C \leq V_{\text{island}}^{\text{max}} \approx 80\text{--}150 \text{ m}^3 \quad (\text{guesstimate for pancaked Shuttle envelope}).$$

7.2 CHARM mass hole

Define island specific power on the **electrical bus**:

$$\alpha_C = \frac{P_\star}{m_C} \quad \Rightarrow \quad \boxed{m_C = \frac{P_\star}{\alpha_C}}.$$

Table 7 CHARM island mass versus specific power at 1 GW.

α_C	m_C at $P_\star = 1 \text{ GW}$	Comment
5 kW/kg	200 t	Heavy; fights SSTO
10 kW/kg	100 t	Stretch
15 kW/kg	67 t	Reference hole

α_C	m_C at $P_\star = 1$ GW	Comment
25 kW/kg	40 t	Aggressive

Volume consistency:

$$\bar{p}_C = \frac{P_\star}{V_C} \Rightarrow V_C = \frac{P_\star}{\bar{p}_C}.$$

For $V_C \leq 120 \text{ m}^3$: $\bar{p}_C \geq 8.3 \text{ MW/m}^3$ bus-averaged over the island—severe packaging.

7.3 Engine and battery holes

$$m_{\text{eng}} = 1.5 \times 10^4 \text{ kg} \quad (\text{single combined-cycle string guesstimate}),$$

$$m_{\text{bat}} = 2.0 \times 10^3 \text{ kg} \quad (\text{restart} + \text{hotel}; \text{ground cart does first light}),$$

$$m_{\text{f}} = 5.0 \times 10^2 \text{ kg} \quad (p\text{-}^{11}\text{B inventory; energy-rich}).$$

7.4 Closed dry / wet mass

$$\begin{aligned} m_{\text{str}} &= m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}, \\ m_{\text{dry}} &= m_{\text{str}} + m_{\text{pl}} + m_{\text{eng}} + \frac{P_\star}{\alpha_C} + m_{\text{bat}} + m_{\text{f}}, \\ m_{\text{w}} &= m_{\text{dry}} \left(e^{\Delta v_{\text{vac}}/v_e} - 1 \right), \\ m_0 &= m_{\text{dry}} e^{\Delta v_{\text{vac}}/v_e}, \\ m_{\text{ins}} &= m_{\text{dry}}. \end{aligned}$$

8 Solved reference vehicle (all-up mass)

Freeze:

$$\begin{aligned} P_\star &= 1 \text{ GW}, \quad \alpha_C = 15 \text{ kW/kg}, \quad \Delta v_{\text{vac}} = 4.0 \text{ km/s}, \\ I_{\text{sp}} &= 2000 \text{ s}, \quad v_e = I_{\text{sp}} g_0 = 19.61 \text{ km/s}. \end{aligned}$$

Solve:

$$m_C = \frac{10^9}{1.5 \times 10^4} = 6.67 \times 10^4 \text{ kg}.$$

$$m_{\text{str}} = 72\,000 + 4\,000 + 3\,000 + 8\,500 = 8.75 \times 10^4 \text{ kg},$$

$$m_{\text{dry}} = 87\,500 + 24\,400 + 15\,000 + 66\,700 + 2\,000 + 500 = 1.961 \times 10^5 \text{ kg} \quad (196.1 \text{ t}),$$

$$\mu = e^{4000/19613} = 1.226, \quad m_{\text{w}} = 0.226 m_{\text{dry}} = 4.43 \times 10^4 \text{ kg} \quad (44.3 \text{ t}),$$

$$m_0 = 2.404 \times 10^5 \text{ kg} \quad (\mathbf{240 \text{ t GLOW}}),$$

$$m_{\text{ins}} = 196.1 \text{ t}.$$

Mass bill (reference):

Table 8 Reference vehicle mass bill.

Item	Mass
Airframe + TPS (longer OML)	72.0 t
Landing gear	4.0 t
Control surfaces + actuators	3.0 t
Crew systems (ECLSS, O ₂ /N ₂ , WCS, galley, food, luggage, airlock)	8.5 t
Payload (cargo bay)	24.4 t
Combined-cycle engine	15.0 t
CHARM island	66.7 t
Flight battery	2.0 t
$p\text{-}^{11}\text{B}$ / proton fuel (+ tankage)	0.5 t
Water (vacuum propellant)	44.3 t
GLOW	240 t

Energy at this mass:

$$E_{\text{orb}} = 6.49 \text{ TJ} = 1800 \text{ MWh}, \quad E_{\text{src}}(\kappa_E = 3) = 5410 \text{ MWh}, \quad t_{\text{src}}(1 \text{ GW}) = 5.4 \text{ h}.$$

Sensitivity (same $\Delta v_{\text{vac}}, I_{\text{sp}}; m_{\text{str}}$ fixed):

Table 9 Sensitivity of dry and wet mass to specific power.

α_C	m_C	m_{dry}	m_w	m_0
10 kW/kg	100 t	229 t	52 t	281 t
15 kW/kg	67 t	196 t	44 t	240 t
25 kW/kg	40 t	169 t	38 t	208 t

Municipal takeoff weight $\sim 210\text{--}280$ t is heavy vs airliners but in the large-military / 747-class band—not a Citation.

9 Constraints on the CHARM power plant

Summarize as a requirement vector $\mathcal{R}_C$:

9.1 Power and mass

$$P_{\text{bus}}(t) \geq P_{\text{prop}}(t) + P_{\text{hotel}}(t), \quad \max P_{\text{bus}} = P_{\star} = 1 \text{ GW},$$

$$m_C = \frac{P_{\star}}{\alpha_C}, \quad \alpha_C \geq 15 \text{ kW/kg (reference; } \geq 10 \text{ kW/kg hard floor for SSTO)}.$$

$$V_C \leq 120 \text{ m}^3, \quad \bar{p}_C = P_{\star}/V_C \geq 8 \text{ MW/m}^3.$$

9.2 Fuel and ash

$$\dot{N}_{p^{11}\text{B}} = \frac{P_{\text{fusion}}}{8.7 \text{ MeV}}, \quad m_{\text{f}}(t_{\text{mission}}) \ll m_{\text{w}}.$$

Ash (He) strained per CHARM multi-chamber design; DEC captures charged-product free energy.

9.3 DEC

$$P_{\text{bus}} = \eta_{\text{DEC}} P_{\alpha, \text{ordered}} + \eta_{\text{th}} P_{\text{thermal reject}}, \quad \eta_{\text{DEC}} \gtrsim 0.4\text{--}0.7 \text{ (design target band)}.$$

Thermal reject (X-ray, wall, inefficiencies) must be dumped to:

- air path (R1–R2 heat exchanger / microwave), and/or
- island bath $\rightarrow$ compact turbine, and/or
- radiators only after exo-atmospheric (limited area).

9.4 Restartable in space

$$E_{\text{restart}} \leq \eta_{\text{bat}} m_{\text{bat}} e_{\text{bat}},$$

with $m_{\text{bat}} = 2 \text{ t}$ reference and $e_{\text{bat}} \sim 0.15\text{--}0.25 \text{ kWh/kg} \Rightarrow 300\text{--}500 \text{ kWh}$ class store—enough for **pilot-chamber relight** + **cascade**, not for ascent.

$$\text{Ground cart: } P_{\text{cart}} \geq P_{\text{light-off}}, \quad \text{not carried.}$$

Doctrine: continuous burn nominally; pilot-string kindling from m_{bat} if segmented; glide if relight fails.

9.5 Continuous operation / no beam farm

Recirculating power is RF walls, rotation, magnets, vacuum, cryo:

$$P_{\text{recirc}} = f_{\text{r}} P_{\text{fusion}}, \quad P_{\text{bus}} = (1 - f_{\text{r}}) P_{\text{fusion, net to bus}}.$$

Design intent: f_{r} small enough that 1 GW **bus** does not require multi-MW neutral-beam injection (NBI).

9.6 Municipal and flight safety

- No tritium breeding inventory.
- Shield so that ramp and cabin doses meet civil constraints with plant running in fan mode.
- Single-string plant: accept engine-out $\equiv$ plant-out $\rightarrow$ glide.

9.7 How CHARM is lit (and how much energy)

CHARM is **not** lit with a neutral-beam farm. Published architecture lights a **rotating open-field mirror** with **species-separated chambers** and **RF / ponderomotive walls** [1,8,9]. A practical light-off sequence for this vehicle is:

Table 10 CHARM light-off sequence.

Step	Action	Power plant elements
0	Evacuate, interlocks, ground-cart connect	Vacuum, controls
1	Energize mirror / chamber magnets	Magnet PSU $\leftarrow$ cart or battery
2	Spin up plasma rotation (centrifugal boron trap)	Rotation drive; energy later recoverable in principle [1]
3	Raise RF one-way / ponderomotive barriers	RF units
4	Inject protons and boron into separated regions	Fuel injectors + p / ^{11}B tanks
5	Establish fusion cell; route alphas to heat-exchange / DEC	DEC electrodes / wave couplers
6	Disconnect cart; bus takes hotel + propulsion	Continuous burn doctrine

Energy scale (engineering estimate — CHARM papers do not publish a flight kWh bill of materials (BOM)):

Magnet + RF + rotation spin-up for a segmented 1 GW island is treated as 50–200 kWh class to first useful plasma (seconds–minutes of MW-class RF/magnet draw), not **MWh-class beams**. That is why

$$E_{\text{restart}} \leq 300\text{--}500 \text{ kWh} \quad (m_{\text{bat}} = 2 \text{ t})$$

is booked for **on-orbit relight**, while **first light on Earth** uses the ground cart. One-way RF walls can be energetically expensive if overused [1]; the vehicle concept of operations (CONOPS) keeps barriers as needed for separation, not as a continuous full-power sink that eats the 1 GW bus.

Pilot-string kindling: battery (or cart) lights a fraction of chambers $\rightarrow$ DEC/RF bus cascades the rest.

10 Combined-cycle engine (detail)

One propulsion string with mode variable $\sigma \in \{1, 2, 3, 4\}$.

10.1 Mode map

Table 11 Combined-cycle engine mode map.

σ	Name	Mach / ambient	Mass flow	Thrust model (0D)
1	Electric fan	$0 \leq M \lesssim 0.9$	$\dot{m}_{\text{air}}$	$T = \eta_f P_{\text{bus}} / v_{\text{jet}}$
2	Thermal ram	$0.9 \lesssim M \lesssim 5$	$\dot{m}_{\text{air}}(M, \rho)$	heat addition $\Delta h = \eta_h P_{\text{th}} / \dot{m}$
3	Plasma / magnetohydrodynamic (MHD)	$M \gtrsim 5$, scarce ρ	small $\dot{m}_{\text{air}}$	$T = \eta_{\text{MHD}} P_{\text{bus}} / v_{\text{eff}}$
4	Water plasma	intakes sealed	$\dot{m}_{\text{w}}$	$T = \dot{m}_{\text{w}} v_e$, $P_{\text{jet}} = \frac{1}{2} \dot{m}_{\text{w}} v_e^2$

Switching laws (schematic):

$$\sigma = \begin{cases} 1 & \rho > \rho_1, M < M_{12}, \\ 2 & \rho > \rho_2, M_{12} \leq M < M_{23}, \\ 3 & \rho_3 < \rho \leq \rho_2, M \geq M_{23}, \\ 4 & \rho \leq \rho_3 \text{ or intakes commanded sealed.} \end{cases}$$

10.2 Fan mode (municipal)

$$P_{\text{fan}} = \eta_{\text{m}}^{-1} T v_{\text{to}}, \quad T \geq (T/W)_{\text{min}} m_0 g_0.$$

For $(T/W)_{\text{min}} = 0.25$, $m_0 = 240 \text{ t}$, $v_{\text{to}} = 80 \text{ m/s}$, $\eta_{\text{m}} = 0.85$:

$$T \approx 0.57 \text{ MN}, \quad P_{\text{fan}} \approx 54 \text{ MW}$$

—well below P_* ; municipal segment is **not** the power driver.

10.3 Ram / plasma modes

Ram drag and net thrust:

$$D_{\text{ram}} = \frac{1}{2} \rho v^2 C_D S, \quad P_{\text{need}} \approx \frac{(D_{\text{ram}} - T_{\text{excess}}) v}{\eta_{\text{p}}}.$$

Peak P_{need} near mid-hypersonic climb sets the 1 GW requirement (design goal G9).

10.4 Water mode

$$\dot{m}_{\text{w}}(t) = \frac{T(t)}{v_e}, \quad \int_0^{t_{\text{vac}}} \dot{m}_{\text{w}} dt = m_{\text{w}},$$

$$P_{\text{bus},4}(t) \geq \frac{1}{2\eta_{\text{jet}}} \dot{m}_{\text{w}} v_e^2.$$

At $T = 100 \text{ kN}$, $v_e = 20 \text{ km/s}$: $P_{\text{jet}} = 1 \text{ GW}$ — vacuum thrust is power-limited; insertion is a **long burn**, not a Space Shuttle Main Engine (SSME) sprint.

10.5 Physical envelope

- One rectangular / axisymmetric nacelle blended to wing root or aft fuselage.
- Variable inlet; **airlock shutters** for R3 and reentry.
- Shared nozzle for modes 2–4; fan bypass duct for mode 1.
- Magnetic bearings / dry lubrication where vacuum survival matters.
- $m_{\text{eng}} = 15\text{ t}$ reference includes motors, microwave/MHD hardware, pumps, and structure.

11 Layout details

Station map for the vision figures in §1. The **top-down bay connectivity** diagram is a reading aid for the floorplan; gear remains on the exterior profile only.

Table 12 Longitudinal station and bay layout.

Station (m)	Bay	Contents
0–11	Crew module	Forward-facing commander (CDR) / pilot (PLT) + flight deck; four more nap-recline seats (six total); WCS; galley; luggage stowage; ECLSS + O₂/N₂ / pressure tanks
—	Ground door	Forward/port crew door (side hatch) — terrestrial ingress only
11–15	Airlock	Suited-crew airlock ($\sim 2.5\text{ m}$ class clear), aft bulkhead facing into cargo bay
15–33.3	Cargo	18.3 m $\times$ 4.6 m payload bay (no reactors)
33.3–35.5	Battery	Flight battery $\approx 2\text{ t}$ (restart / hotel)
35.5–37.5	Fusion fuel	Proton / ¹¹ B feed tanks + plumbing (low mass, real volume)
37.5–45	CHARM	Reactor island ($\lesssim 120\text{ m}^3$, 67 t)
45–49	Water	$\approx 44\text{ t H}_2\text{O}$
49–52	Engine	Combined-cycle nacelle + nozzle
Wings	Controls	Elevons, rudder (gear not drawn on this figure)

Doors (Shuttle pattern). (1) **Side/forward crew door** — runway/ground only; (2) **airlock** — on-orbit cabin $\leftrightarrow$ cargo bay / vacuum for suited operations [21].

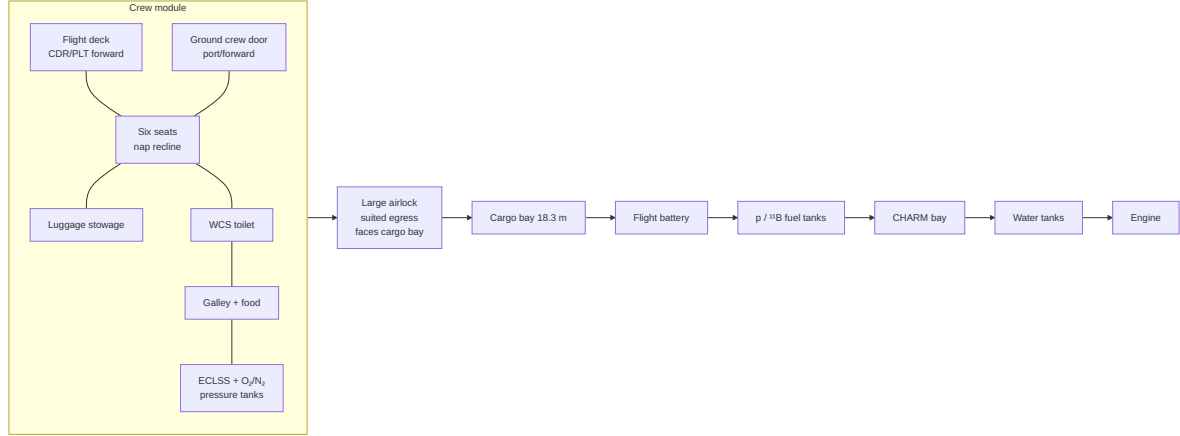


Figure 5 Top-down floorplan.

12 Systems checklist (equations → constraints)

Table 13 Systems checklist: equations to constraints.

Step	Governing relations	Binding output
Goals	G1–G9 + single-deck crew	Shuttle bay + municipal SSTO to ISS
LEO energy	$E_{\text{orb}} = m_{\text{ins}}\Delta\varepsilon$, $E_{\text{src}} = \kappa_E E_{\text{orb}}$	~ 1.8 GWh orbital @ 196 t; ~ 5.4 GWh source @ $\kappa_E = 3$
Regimes	R1–R3 mass-flow logic	Free air → scarce air → water
Water	$m_w = m_{\text{dry}}(e^{\Delta v_{\text{vac}}/v_e} - 1)$	$\sim 23\%$ of dry mass @ ref
Structure	$m_{\text{str}} = m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}$	87.5 t incl. gear/controls/ECLSS+O ₂ /luggage/airlock
CHARM	$m_C = P_{\star}/\alpha_C$, DEC, restart	67 t @ 15 kW/kg
Light-off	Magnets + RF + rotation; no NBI	50–200 kWh est.; cart / 2 t battery
Solve	$m_0 = m_{\text{dry}}e^{\Delta v_{\text{vac}}/v_e}$	$m_0 \approx 240$ t; $L \approx 52$ m
Engine	Mode map σ ; $P_{\text{fan}} \ll P_{\star}$; $P_{\text{jet}} = \frac{1}{2}Tv_e$	1 GW sizes R2–R4

13 Imputed CHARM plant specs, gap to present design, and unobtainiums

13.1 Specs this vehicle imputes to CHARM

The SSTO solve does not invent new plasma physics; it **back-solves** a power island that must fit the airframe. Reference imputed plant:

Table 14 Imputed CHARM plant requirements for this SSTO.

Quantity	Imputed requirement	Source in this note
Fuel	Continuous $p\text{-}^{11}\text{B}$	G8; §2
Architecture	Multi-chamber rotating mirror; species separation; ash strain; DEC	§3, §9, §1; [1,8–11]
Electrical bus peak	$P_\star = 1\text{ GW}$	G9; Phase B/C power
Mission source energy	$\sim 5\text{ GWh}$ class per ascent ($\kappa_E \sim 3$)	§4, §8
Island mass	$m_C \approx 67\text{ t}$	$\alpha_C = 15\text{ kW/kg}$
Specific power (bus)	$\alpha_C \geq 15\text{ kW/kg}$ (floor $\sim 10\text{ kW/kg}$)	§7.2, §9.1
Island volume	$V_C \lesssim 120\text{ m}^3$	Fuselage bay aft of cargo
Volumetric power	$\bar{p}_C \gtrsim 8\text{ MW/m}^3$	$P_\star/V_C$
DEC	$\eta_{\text{DEC}} \sim 0.4\text{--}0.7$ on ordered α / wave channel	§9.3
Aux heating	RF + rotation + magnets; no multi-MW NBI farm	§9.5, §9.7
Light-off energy	$\sim 50\text{--}200\text{ kWh}$ to useful plasma (est.)	§9.7
Space restart	Pilot-string from $\sim 2\text{ t}$ battery ($\sim 300\text{--}500\text{ kWh}$)	§6, §9.4
Duty	Continuous burn through climb/insert; throttleable bus	§5
Environment	Flight loads, TPS-adjacent thermal, municipal dose with fan-mode plant	G6, §9.6

13.2 Where published CHARM stands today

Relative to that invoice, the Fisch / Advanced Research Projects Agency–Energy (ARPA-E) / Pale Blue line today is a **physics and intellectual-property (IP) program**, not a flight power plant [1,2]:

Table 15 Gap between present CHARM and vehicle need.

Gate	Present CHARM (public)	Vehicle need
Fuel / kinetics	Strong papers on hybrid fast–thermal $p\text{-}^{11}\text{B}$, alpha channeling, ash poisoning [3–7]	Same fuel family — aligned in intent
Chambering	Architecture + patent filings for separated reactants, ponderomotive walls, open-trap HV, differential confinement [1,8–11]	Must work together in one island
Lawson / Q	Component studies and 0D balances; integrated self-consistent power-positive reactor not demonstrated [1]	Net bus power after recirculating RF/rotation
Hardware	No public GW-class (or even pilot) CHARM machine; software-first / early company path	Flyable 67 t island
DEC	Themes in theory and open-trap high-voltage (HV) patents [1,10]	Flight-qualified η_{DEC} at GW

Gate	Present CHARM (public)	Vehicle need
Mass / volume	Not published as kW/kg or MW/m ³ plant envelopes	15 kW/kg, $\gtrsim$ 8 MW/m ³
Operations	Lab/site thinking	Continuous ascent, g-load, restart, airport licensing

Gap in one line: CHARM has a credible **aneutronic architecture story**; this vehicle needs a **closed, flight-packaged, gigawatt, high specific-power product** that does not yet exist on paper as an engineered BOM.

13.3 Unobtainiums in the gap

Call a requirement **unobtainium** if it is not implied by present CHARM results and would break the SSTO close if false. Ranked for this airframe:

1. **Integrated p -¹¹B power balance with chambered species + ash removal + tolerable synchrotron/bremsstrahlung simultaneously** — components suggest feasibility; full self-consistency is explicitly still open [1].
2. **Centrifugal / rotating-mirror differential confinement at useful density and confinement time**, including rotation sustainment **without intolerable wall voltage drops** [1].
3. **Selective RF / ponderomotive “one-way” walls** that regulate ion traffic at acceptable recirculating power (slides note one-way walls can be energetically costly if overused) [1,9].
4. **Wave-mediated ash extraction / alpha channeling into protons (or DEC)** fast enough that helium does not poison the cell [3,6].
5. **Ultra-high DC / open-field electrode structures** that survive α and X-ray loads while feeding a GW bus [10].
6. **Flight specific power** $\alpha_C \sim 15 \text{ kW/kg}$ **and** $\bar{p}_C \gtrsim 8 \text{ MW/m}^3$ including magnets, RF, shield, cryo, and structure — beyond any published CHARM packaging study.
7. **Continuous GW-class operation** through a multi-hour ascent with vibration, thrust-vector loads, and thermal transients — not part of the present ARPA-E scope.
8. **Pilot-string light-off / space restart** at 50–200 kWh class — engineering estimate only; not a CHARM experimental result.

Items 1–5 are **physics/architecture unobtainiums** shared with any CHARM plant. Items 6–8 are **aerospace packing and CONOPS unobtainiums** imposed by SSTO. If 1–5 fail, no amount of airframe cleverness saves the mission; if 1–5 hold but 6–8 fail, CHARM may still be a ground plant while this spaceplane does not close.

13.4 How to read the rest of this paper

Sections 1–12 are a **requirements mirror** held up to CHARM: they say what a successful bottle must look like to fly Shuttle-class cargo from a municipal runway to ISS altitude. They are **not** a claim that Pale Blue / Princeton has those numbers. Closing the gap is future plasma physics, materials, and packaging work—tracked against the unobtainium list above.

14 Conclusion

Design goals fix a **Shuttle-style SSTO** with a **real cargo bay**, municipal runway ops, and ISS-altitude LEO. The pressurized nose is **lengthened** (≈ 52 m **OML**) so the cabin can hold a real **ECLSS with O₂/N₂, luggage**, a **suited-size airlock** into the bay, plus visible **battery** and **fusion-fuel tank** bays. Crew: **forward-facing flight deck**, **six nap-recline seats**, **ground crew door**, **aft airlock into cargo**. **Landing gear** (4 t) and **control surfaces** (3 t) sit in the mass bill (gear drawn on the profile, not the floorplan). The MWh budget is linear in inserted mass: about 9.2 kWh **per kg** orbital, times $\kappa_E \sim 2\text{--}4$ at the CHARM bus. Water scales only with **dry mass** and vacuum $\Delta v/I_{sp}$. Closing $P_\star = 1$ GW at $\alpha_C = 15$ kW/kg yields a reference vehicle of about 196 t **dry**, 44 t **water**, 240 t **GLOW**, with CHARM as a 67 t, $\lesssim 120$ m³ island using **DEC**, lit by **magnets + RF + rotation** (50–200 kWh class light-off; space restart from a 2 t battery), and a **single combined-cycle engine** spanning fan $\rightarrow$ ram/plasma $\rightarrow$ water jet. Vision figures (§1) show planform, plant, and profile. Those are the numbers the in-silico CHARM plant and engine maps must satisfy.

15 Acknowledgments

CHARM denotes the chambered aneutronic rotating-mirror architecture developed at Princeton Plasma Physics Laboratory (PPPL) under the ARPA-E economical $p\text{--}^{11}\text{B}$ program [1]–[11] and discussed toward Pale Blue Fusion. This vehicle sketch is an independent systems exercise and does not speak for that program.

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